



# Engineering the facilities advanced industry runs on.

EPC, Process Critical Utilities & Total Tool Installation Solutions, and Energy Management. For cleanroom-critical industries: semiconductor, data centre, EV battery, pharmaceutical and energy. Established 1995.

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# 01

## The Company

SECTIONS 01.1 TO 01.9

Established 1995 · 31 years in total facility solutions

450 people across 7 global offices

ISO 9001 · ISO 14001 · ISO 45001 certified



## About IAQ

Who we are and what we build. The delivery models are in 01.7.

# Established 1995. A trailblazer in total facility solutions.

Engineering, procurement, construction and maintenance for the industries where contamination, uptime and safety decide the outcome, from semiconductor wafer fabs to gigafactories, data centres and pharmaceutical plants.

### Vision

To be a regional facility solutions provider with engineering excellence that facilitates technological innovation and advancement in quality of life.

### Mission

Providing innovative, sustainable facility and engineering solutions benefitting our clients and stakeholders, driven by our leadership, employees and partners globally.

## IAQ at a Glance

The company in six numbers, and the certifications that stand behind them.



 **31**  
YEARS EXPERIENCE

 **450**  
EMPLOYEES

 **200+**  
PROJECTS COMPLETED

 **1.5 mil m<sup>2</sup>**  
CLEANROOM BUILT-UP AREA

 **7**  
GLOBAL OFFICES

 **20 t / yr**  
CARBON FOOTPRINT REDUCED

### CERTIFICATION & RECOGNITION

INTERTEK **ISO 9001:2015**

INTERTEK **ISO 14001:2015**

INTERTEK **ISO 45001:2018**

CIDB **GRADE G7**

PKK **GRADE G7**

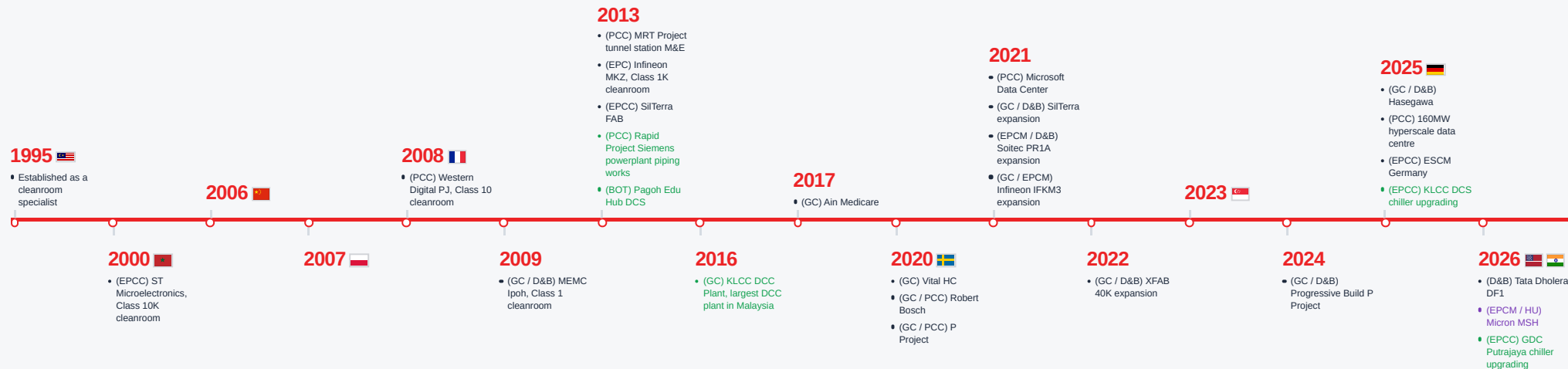
HIGHWIRE SAFETY **GOLD AWARD**

MCIEA 2024 **BUILDER OF THE YEAR**

# Milestones

IAQ's own timeline, 1995 to 2026: the projects that marked each year and the offices opened along the way.

● EPCC PROJECTS ● ENERGY PROJECTS ● TOOL UTILITY PROJECTS ● OFFICE OPENED



## Global Footprint

Seven markets, from the Malaysian headquarters outward.



**1995**

### Malaysia

Turnkey design, procurement, construction and commissioning for hi-tech clients. Selangor (HQ), Penang, Johor and Kuching.



**2020**

### Sweden

Gigafactory construction for EV battery manufacture, part of Europe's clean and digital transition.



**2023**

### Singapore

Design & build EPCM services to global clients in the semiconductor industry.



**2025**

### Germany

EPCC services following the semiconductor industry expansion in Europe.



**2026**

### India

D&B turnkey delivery for India's first wafer fab.



**2026**

### United States

Entry to the US following the localisation of advanced technology facilities.



—

### Ireland

Newest market. Scope and year to be confirmed by IAQ.

## Core Values

Six commitments that govern how every project is run. Not words, but working rules.



01

### Safety First

Our services and works are carried out to the highest standard of safety and ethics.



02

### Quality Consistency

We take pride in the quality of solutions delivered, to achieve maximum client satisfaction.



03

### Honesty and Integrity

Our core code of conduct, fostering trust, accountability, professionalism and ethical practice.



04

### Efficiency & Proficiency

Committed to maximising available resources to achieve the best result collectively.



05

### Engineering Capabilities

Our way of working revolves around engineering principles, developing precise and sustainable solutions.



06

### Pursuit of Excellence

Devoted to excellence in all our works, providing sustainable solutions to complex challenges.

## Business Units

Three delivery capabilities under one group, covering the full chain.



### 01

#### Engineering, Procurement and Construction (EPC)

EPC from conception to operation, completion and maintenance. Every stage managed, from initial design through commissioning, so clients in hi-tech industries can bring their visions to life.



### 02

#### Process Critical Utilities & Total Tool Install Solutions

EPCM partner for semiconductor manufacturing: process-critical utility infrastructure and total tool installation, bridging facility readiness and manufacturing start-up to accelerate fab ramp-up.



### 03

#### Energy Management

Energy management solutions that optimise operations and reduce carbon footprint, keeping facilities running at the highest levels of efficiency and sustainability.

## Delivery Models

The three models IAQ contracts under, across the full life of a facility.

### EPCM

#### Engineering, Procurement & Construction Management

Management of engineering, procurement and construction for semiconductor manufacturing, bridging facility readiness and manufacturing start-up to accelerate fab ramp-up.

##### DELIVERED UNDER THIS MODEL

- SilTerra expansion
- Soitec PR1A expansion
- Infineon IFKM3 expansion
- Micron MSH

### EPCC

#### Engineering, Procurement, Construction & Commissioning

Full turnkey delivery from conception to operation, completion and maintenance, with every stage managed from initial design through commissioning.

##### DELIVERED UNDER THIS MODEL

- ST Microelectronics
- SilTerra FAB
- ESCM Germany
- KLCC DCS chiller upgrading

### EFM

#### Energy Facility Management

Ongoing energy and facility management that optimises operations and reduces carbon footprint, keeping facilities at the highest levels of efficiency.

##### DELIVERED UNDER THIS MODEL

- KLCC DCC Plant
- GDC Putrajaya chiller upgrading
- Pagoh Edu Hub DCS

## Scope of Services

Eight packages, self-performed and integrated under one contract.



01

### Civil, Structural & Architectural

Greenfield and brownfield CSA



02

### Air Conditioning System

ACMV, make-up air, recirculation



03

### HT & LV Electrical System

Up to 33kV substations



04

### Fire Protection System

Detection, suppression, compliance



05

### Cleanroom System

ISO 3 to ISO 8 · Class 1 to Class 100K



06

### Process Utilities

CDA, PCW

- Process Exhaust System
- Chemical / Gas Delivery System
- Ultra Pure Water (UPW)
- Waste Water Treatment



07

### Facility Monitoring & Control System

FMCS, BMS, PA



08

### Tools Hookup

Progressive tool install and hookup

## Industry Focus

Seven sectors, one discipline: contamination control under code and class.



### Semiconductor

Wafer fab, advanced packaging, test & assembly



### Data Centre

Hyperscale cooling, security, uptime



### EV Battery

Dry rooms, moisture control, explosion proofing



### District Cooling & Heating

Urban energy plant and distribution



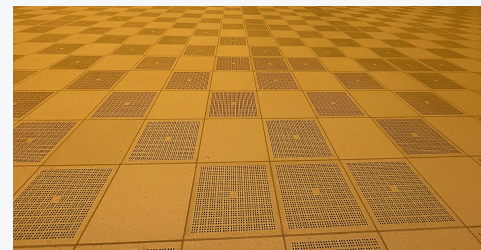
### Photovoltaics

Cell and module lines, toxic material handling



### Pharmaceuticals & Hospitals

GMP grades, regulated environments



### Food & Beverage

Hygienic design, quality and safety standards



## Your facility next

Total facility solutions, end to end.

PART 02 OF 03

# 02

## Project References

SECTIONS 02.1 TO 02.11

200+ projects completed across three regions

Over 1.5 million m<sup>2</sup> of cleanroom built

ISO 3 to ISO 8 · Class 1 to Class 100K



## Portfolio Overview

How the reference book breaks down by region, and which sections cover what.



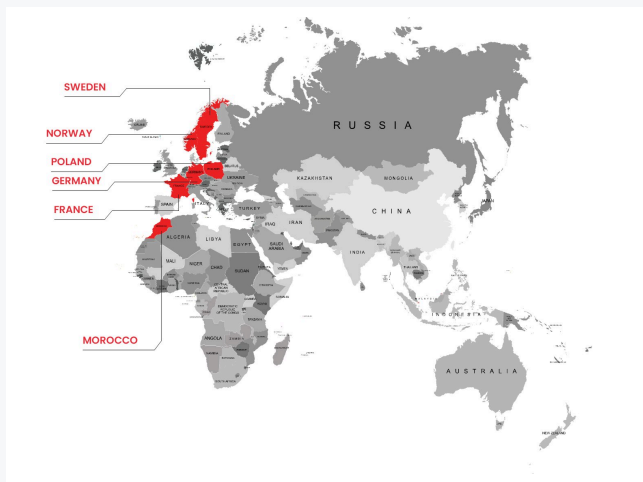
02.2 TO 02.4 · 02.8 TO 02.10

### Malaysia & Singapore

Semiconductor, data centre, pharmaceutical, photovoltaics, district cooling and energy.

**200+**

PROJECTS COMPLETED



02.5 TO 02.6

### Europe & Morocco

EV battery gigafactories, wafer fabs and advanced semiconductor in Sweden, Norway, France, Poland and Morocco.

**1.5 mil m<sup>2</sup>**

CLEANROOM BUILT



02.7

### China

Semiconductor, display, pharmaceutical, automotive and R&D facilities.

**5**

CONTRACT MODELS

GC · EPCC · EPCM · PCC · Tool Hookup

# Semiconductor · Malaysia

Wafer fab, advanced packaging, test and assembly. Further Malaysian projects are listed in 02.11.



KULIM, KEDAH · GREENFIELD WAFER FAB

## Infineon Kulim Wafer Fab 3

|             |                                                                       |
|-------------|-----------------------------------------------------------------------|
| DESCRIPTION | General contractor for WP06, KLM3 Expansion, 35,000 m² cleanroom area |
| SCOPE       | PCC for cleanroom and mechanical works                                |
| CLEANROOM   | ISO 4 · 5 · 6 · 7 (CLASS 10 · 100 · 1K · 10K)                         |



ADVANCED PACKAGING · 75,000 M²

## P Project

|             |                                                                                          |
|-------------|------------------------------------------------------------------------------------------|
| DESCRIPTION | Progressive-built EPCM contract model for the client's advanced packaging facility       |
| SCOPE       | Design & build GC: CSA, MEP, process utilities, chemical & gases system, waste treatment |
| CLEANROOM   | ISO 4 TO 7 (CLASS 10 TO 10K)                                                             |

Semiconductor · Singapore

Design & build EPCM in Singapore since 2023.



PHOTOMASK FACILITY

Teksend Photomask

|             |                                                                                                           |
|-------------|-----------------------------------------------------------------------------------------------------------|
| DESCRIPTION | Addition and alteration to an existing two-storey building with cleanroom facilities and ancillary office |
| SCOPE       | EPCC for cleanroom, ACMV, electrical, process utility, BMS, PA, WWT, UPW, tool hookup                     |
| CLEANROOM   | CLASS 1 TO CLASS 10K                                                                                      |



PROGRESSIVE TOOL HOOKUP

Micron MSH Tool HU

|             |                                                |
|-------------|------------------------------------------------|
| DESCRIPTION | 2,000 progressive tool hookup services for MSH |
| SCOPE       | Progressive tool hook up                       |
| CLEANROOM   | Not applicable                                 |

## Data Centre

Hyperscale and colocation, 9.6 MW to 160 MW.



CLIENT CONFIDENTIAL · 160 MW

### Hyperscale Data Centre

|             |                                                                                 |
|-------------|---------------------------------------------------------------------------------|
| DESCRIPTION | Office, data halls and mechanical utility building                              |
| SCOPE       | Construction, completion, testing & commissioning for mechanical packages 1 & 2 |
| CLEANROOM   | Not applicable                                                                  |



KUL03 PHASE 1 · 9.6 MW

### Microsoft DTC-KUL 03

|             |                                 |
|-------------|---------------------------------|
| DESCRIPTION | 9.6 MW data centre              |
| SCOPE       | PCC for CHW and CW piping works |
| CLEANROOM   | Not applicable                  |

# EV Battery · Europe

Dry rooms for the gigafactories: moisture control and contamination control at scale.



SKELLEFTEÅ, SWEDEN · 62,000 M<sup>2</sup>

## Northvolt AB Gigafactory

|             |                                                        |
|-------------|--------------------------------------------------------|
| DESCRIPTION | Europe gigafactory for lithium-ion battery manufacture |
| SCOPE       | Cleanroom and dry room architecture works              |
| CLEANROOM   | Not applicable                                         |



NORWAY · BATTERY COAST · 8,600 M<sup>2</sup>

## Morrow Batteries

|             |                                                   |
|-------------|---------------------------------------------------|
| DESCRIPTION | Norway's Battery Coast project                    |
| SCOPE       | Cleanroom and dry room architectural system, ACMV |
| CLEANROOM   | Not applicable                                    |

## Semiconductor · Europe & Morocco

Following the fabs into Europe: advanced-node cleanrooms and progressive hookup.



CLIENT CONFIDENTIAL · WAFER FAB

### 28/22nm CMOS · 16/12nm FinFET Plant

DESCRIPTION Design, supply, installation & commissioning of cleanroom facilities

SCOPE Cleanroom & mechanical general contractor

CLEANROOM CLASS 1 TO CLASS 10K



CASABLANCA, MOROCCO · 60,000 M²

### STMicroelectronics

DESCRIPTION Built-up area up to 60,000 m²

SCOPE Cleanroom, mechanical, electrical, process utilities and hookup works

CLEANROOM ISO 5 (CLASS 100)

# Semiconductor & Display · China

EPCC across semiconductor, display and precision manufacturing.



HANGZHOU · 23,400 M<sup>2</sup>

## Ferrotec Semiconductor

|             |                                                                          |
|-------------|--------------------------------------------------------------------------|
| DESCRIPTION | Built-up area up to 23,400 m <sup>2</sup>                                |
| SCOPE       | EPCC for cleanroom, mechanical, electrical and plumbing works and hookup |
| CLEANROOM   | ISO 4 · 5 (CLASS 10 & 100)                                               |



WUXI · 8,000 M<sup>2</sup>

## Infineon Technologies

|             |                                                                          |
|-------------|--------------------------------------------------------------------------|
| DESCRIPTION | Built-up area up to 8,000 m <sup>2</sup>                                 |
| SCOPE       | EPCC for cleanroom, mechanical, electrical and plumbing works and hookup |
| CLEANROOM   | ISO 5 (CLASS 100)                                                        |

# Photovoltaics

Solar cell and module manufacturing, including toxic material and waste handling.



KULIM, MALAYSIA

## First Solar Malaysia

|             |                                                                                                |
|-------------|------------------------------------------------------------------------------------------------|
| DESCRIPTION | First Solar KMW building in Kulim                                                              |
| SCOPE       | PCC for mechanical & electrical, KMW building; CSA and M&E for forming gas plant; M&E for SCO1 |
| CLEANROOM   | Not applicable                                                                                 |



CHINA · 57,000 M<sup>2</sup>

## Chint Solar Technology

|             |                                                            |
|-------------|------------------------------------------------------------|
| DESCRIPTION | Built-up area up to 57,000 m <sup>2</sup>                  |
| SCOPE       | Cleanroom system, ACMV, process utilities and tools hookup |
| CLEANROOM   | ISO 8 (CLASS 100K)                                         |

# Pharmaceutical & Medical

GMP grades and regulated environments, from parenteral facilities to medical device plants.



GREENFIELD · MEDICAL DEVICE

Insulet

|             |                                                                   |
|-------------|-------------------------------------------------------------------|
| DESCRIPTION | Electronic medical device manufacturing plant, greenfield project |
| SCOPE       | Design and build mechanical, cleanroom & electrical package       |
| CLEANROOM   | ISO 8 (CLASS 100K)                                                |



PUCHONG, SELANGOR

Pharmaniaga

|             |                                                               |
|-------------|---------------------------------------------------------------|
| DESCRIPTION | Small volume parenteral facility                              |
| SCOPE       | Cleanroom system, air conditioning and mechanical ventilation |
| CLEANROOM   | ISO 5 · 7 · 8 (CLASS 100 · 10K · 100K)                        |

# District Cooling & Energy

Plant, chillers and co-generation for urban and industrial energy.



GAS DISTRICT COOLING (M) SDN BHD

## KLCC District Cooling Plant

DESCRIPTION LARGEST DISTRICT COOLING CENTRE IN MALAYSIA

SCOPE ACMV, electrical & instrumentation, fire protection, process utility, plumbing & sanitary, water treatment, CSA

CLEANROOM Not applicable



PICC PLANT, PUTRAJAYA

## GDC Putrajaya

DESCRIPTION For the Putrajaya International Convention Centre (PICC) plant

SCOPE EPCC of plant electrification, chiller replacement and associated works at Gas District Cooling Putrajaya

CLEANROOM Not applicable

## Reference Index

A selection from 200+ completed projects. The full reference list is available on request.

### MALAYSIA · SEMICONDUCTOR

XFAB Kuching 40K Expansion, 40,000 m²

SOITEC PR1A Expansion, 45,000 m²

Robert Bosch Testing Manufacturing Plant

Wafer Fab Facility & Expansion, FAB1E A/B

Texas Instruments, Melaka, 32,000 m²

Infineon Technologies, Melaka, Block 8

MEMC Ipoh, Class 1 cleanroom

STMicroelectronics, Johor

Western Digital (M), Petaling Jaya

WD Media (M), Penang, MA 28 & MA 29

Flextronics Shah Alam

### MALAYSIA · OTHER SECTORS & SINGAPORE

AUO Sunpower, Melaka, Fab 3A

Vital Healthcare, medical device plant

Ain Medicare, pharmaceutical building

M Project, food flavouring plant, 40,000 m²

P Project, MEP works, mega lab retrofit

Rapid Petronas Pengerang Co-gen

Pagoh Education Hub DCS

Rapid MCD chemical plant

Matrix (M) Sdn Bhd

T Hasegawa, spray dryer & microwave tunnel

Elegant Aura (M) Sdn Bhd

Caterpillar Asia, 7T office renovation

Micron F10A & F10NX, Singapore

National University of Singapore, tissue culture lab

### EUROPE & MOROCCO

ACC Phase 1, France, 35,000 m²

Envision, France

Soitec Semiconductor, Paris, 11,000 m²

Xfab Semiconductor, Paris

Nemotek, Rabat, Morocco

Kimoto, Poland

Sumika, Poland

### CHINA

SMIC, Ningbo, 22,000 m²

Maxtor Technology, Suzhou, 95,000 m²

All-Cent RF Technology, Wuxi, 25,600 m²

Kunshan Visionox Display, 27,000 m²

GS Magicdrive Inc., 68,000 m²

Shantou Goworld Display

M-Flex (Suzhou), 25,000 m²

MMI Industries (Wuxi)

Huawei Technologies Co., Ltd

Shanghai Roche Pharmaceuticals

Suzhou Ascentage Pharma, 25,800 m²

Jiangsu GenScript Biotech

Mindray Group Co., Ltd

Unilever · GE · Dow · Rohm and Haas R&D centres

PART 03 OF 03

# 03

## Safety, Quality & ESG

SECTIONS 03.1 TO 03.4

Highwire Gold safety award

ISO 45001:2018 occupational health & safety

20 tons per annum carbon footprint reduction



## ESG Commitments

Three pillars, applied to operations and to every project we take on.



### E

#### Environmental Responsibility

Minimising environmental footprint through sustainable design and construction. Energy-efficient solutions and eco-friendly technologies, prioritised at design stage.



### S

#### Social Accountability

The well-being and safety of employees and communities first. A diverse, inclusive work environment and ethical labour practices.



### G

#### Corporate Governance

Transparency, accountability and ethical decision-making, with the highest standards of compliance, building trust with clients, partners and stakeholders.

## Safety & Recognition

How our standards are independently verified and recognised.

# Judged on the things that actually matter.

MCIEA award recipients are evaluated across company performance, project management, technical expertise, innovation, quality, safety and sustainability. The same criteria our clients audit us against.

### Builder of the Year 2024

Malaysian Construction Industry Excellence Awards (MCIEA), awarded by CIDB.

### Highwire Gold

Gold-rated contractor safety performance.

### ISO 45001:2018

Occupational health and safety management, certified by Intertek.



## Closing

Your Total Facility Solutions Provider.

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# We infuse safety, quality, efficiency and excellence into every project.

From engineering design and project management to construction and maintenance, across semiconductor manufacturing, clean energy and every environment in between. IAQ is your trusted partner for total facility solutions.

## Contacts

For inquiry and business opportunity.



YOUR TOTAL FACILITY SOLUTIONS PROVIDER

Engineering, procurement, construction and energy management for the industries where contamination, uptime and safety decide the outcome. Established 1995.

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